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SmartGlycoAI

Stage 1: Project proposal



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SmartGlycoAI

Project Proposal - Stage 1

Group members

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Problem description

People with diabetes who take insulin have to think about their treatment every time they eat. Before each meal they need to estimate how many carbohydrates are in the food and then calculate how much insulin to take. This happens several times a day, at home, at work or in a restaurant. Most importantly, only guessing carbohydrates just by looking at the food is difficult, and doing the calculations every time is tiring and leads to mistakes.

The information about their illness is also spread across different places. Their glucose values are in a sensor app. Meals and doses may be in their memory or in a paper notebook. This makes it harder to understand how food affects their glucose over the week. When they go to an appointment, a doctor or nurse may get incomplete information about the patient's cycle and have to rely on what they remember.

Target users

The main users are people with diabetes type 1 who use insulin, of different ages and with different levels of experience with technology. Many of them already use a glucose sensor (CGM). The secondary users are doctors and nurses who follow these patients and want to see their history.

Project goal

The goal of the project is to make daily diabetes management easier and with fewer errors, and to keep the patient and the healthcare team connected. A possible solution is a mobile app where the user can:

1. Take a photo of a meal so the app identifies the food, shows the carbohydrates and suggests an insulin dose based on the user's own settings. The user checks and confirms before it is saved.
2. See the current glucose level from the sensor and a graph of the last 7 days on the home screen.
3. Look at a diary with the history of meals and corrections, and manage their profile, settings and devices.
4. Share their data with their doctor or nurse.

Competing application

SNAQ is an app that estimates the carbohydrates of a meal from a photo and connects to glucose sensors to show how each meal affected the glucose level. The positive aspects are that it is fast and reduces mistakes in counting carbohydrates. The negative aspects are that photo carb counting needs a paid subscription, it does not suggest an insulin dose so the user still has to calculate it, it has many extra features that make the main task less simple, and the doctor only receives exported reports.

Our app focuses on a simple flow (photo, carbohydrates, dose, diary) and gives healthcare professionals direct access to the data.